appendix A. UML blockchain design models

Software application development should always begin with a clear problem statement that describes a problem to be solved, including its requirements, scope, limitations, exceptions, and expected outcomes. You analyze this problem statement to come up with a design representation. The design representation of an application is like the blueprint that is created before the construction of a home or the engineering design created before a product is machined.

Software application developers are often eager to jump into coding before they design, but this is not good practice. The best practice is to analyze and design the solution to a problem in a standard format so that all the parameters can be discussed with the stakeholders in an implementation-independent fashion, using visual representations of the design components. Unified Modeling Language (UML) (https://www.uml.org) offers multiple diagram models for design representation.

The UML design methodology was introduced about three decades ago to address the development challenges that arose as the scale and complexity of software increased, with small-scale systems being replaced by large, multimodule systems. UML modeling has been widely